

MASTER ENGINEERING E-CATALOGUE

Industrial Compressed Air & Process Thermal Systems

Engineered for extreme ambient conditions and rigorous 24/7 industrial duty. Precision-manufactured in Arasur, Coimbatore, serving textile mills, pharmaceutical plants, automotive manufacturing, plastic injection molding, and heavy precision engineering nationwide.

Air Treatment Systems

Refrigerated Dryers (WRD, 20–2000 CFM), Heatless Desiccant Dryers (WHD, -40°C PDP), In-Line Microfilters (WMF), and Zero Air Loss Automatic Drains (WADV).

Thermal Management

Air & Water-Cooled Industrial Chillers (WCP, 2–100 TR), Specialized Anodizing & Medical Units, and Industrial Flake Ice Plants (WFI, 1–30 TPD).

Heat Rejection Plant

Aerodynamic FRP Round Bottle & Square Crossflow Cooling Towers (WCT, 10–1500 TR) and Closed-Circuit Coil Towers (WCC, 10–500 TR).

CORPORATE PROFILE & QUALITY SYSTEMS

About Win Equipments

Leading Indian Manufacturer of Heavy-Duty Compressed Air Treatment & Thermal Process Engineering

Precision Manufacturing Infrastructure in Coimbatore: Win Equipments operates an advanced manufacturing facility at Arasur, Coimbatore, equipped with precision CNC sheet metal fabrication, ASME-compliant vessel welding cells, automated coil expanding, and cleanroom refrigeration brazing.

With decades of hands-on application engineering across tropical Indian climates, our machinery is built to withstand high ambient temperatures (+45°C), power quality fluctuations, and heavy airborne lint and dust. Every refrigeration circuit undergoes hydrostatic proof testing, dual helium mass-spectrometer leak detection, and an 8-hour continuous load trial prior to shipment.

ARASUR PLANT FACILITY

ISO 9001:2015

ENERGY-EFFICIENT DESIGNS

PAN-INDIA FIELD SUPPORT



CORE INDUSTRIAL CAPABILITIES & STANDARDS

ASME & IS Pressure Vessels

In-house welding to ASME Section VIII Div 1 and IS 2825 Class I/II standards with 100% radiographic weld seam testing and 1.5x hydrostatic qualification.

Eco-Friendly Refrigerants

Zero ODP refrigerants including R134a, R407C, and R410A paired with high-efficiency Copeland scroll and Bitzer semi-hermetic reciprocating/screw compressors.

Microprocessor Automation

Digital micro-controllers and PLC automation panels with RS-485 Modbus RTU communication for seamless DCS/SCADA plant integration.

MAJOR INDUSTRIES SERVED

Textile Spinning & Weaving

Automotive OEMs & Ancillaries

Pharmaceutical & API Formulation

Plastic Injection & Blow Molding

Food, Dairy & Cold Storage

Chemical & Electroplating Plants

Laser Cutting & Metal Fabrication

AIR TREATMENT • WRD SERIES (20 TO 2000 CFM)

Refrigerated Compressed Air Dryers

High-Efficiency Non-Cycling Direct Expansion Dryers Delivering Constant +3°C Pressure Dew Point

The WRD Series combines an air-to-air precool regenerator, direct-expansion evaporator, and cyclonic moisture separator in a 3-in-1 modular exchanger. Rated for inlet temperatures up to 50°C and pressures from 4.0 to 16.0 bar g with pressure drops below 0.2 bar.

CONSTANT +3°C DEW POINT

20-2000 CFM

ZERO AIR LOSS DRAIN

MAX 16 BAR G



WRD SERIES TECHNICAL SPECIFICATION MATRIX

MODEL	CFM	m ³ /hr	kW	Connection	Dimensions (mm)	Weight
WRD 20 S	20 CFM	34	0.45 kW	1/2" BSP	450 × 450 × 550	38 kg
WRD 60 S	60 CFM	102	0.65 kW	1" BSP	500 × 550 × 750	55 kg
WRD 100 S	100 CFM	170	0.85 kW	1" BSP	500 × 550 × 750	60 kg
WRD 200 S	200 CFM	340	1.25 kW	1.5" BSP	600 × 650 × 900	95 kg
WRD 300 S	300 CFM	510	1.65 kW	2" BSP	600 × 650 × 900	110 kg
WRD 500 S	500 CFM	850	2.20 kW	2.5" BSP	800 × 800 × 1100	190 kg
WRD 1000 S	1000 CFM	1700	5.50 kW	4" FFlange	1500 × 1000 × 1500	450 kg
WRD 2000 T	2000 CFM	3400	10.5 kW	6" FFlange	1800 × 1200 × 1600	750 kg

Hot Gas Bypass Control: Modulates refrigeration capacity automatically from 0% to 100% pneumatic flow, guaranteeing freeze protection and continuous dew point stability without compressor start/stop cycling.

AIR TREATMENT • WHD SERIES (300 TO 2000 CFM)

Heatless Twin-Tower Desiccant Air Dryers

Pressure Swing Adsorption (PSA) Technology Providing -40°C to -70°C Sub-Zero Pressure Dew Point

Engineered for sensitive instrumentation, pharmaceutical cleanrooms, robotic painting, and semiconductor fabrication. Twin carbon steel vessels filled with high-crush-strength activated alumina operate on an automated 8-minute PSA cycle with low purge loss (12–15%).



-40°C STANDARD DEW POINT PSA CYCLE ACTIVATED ALUMINA 4 TO 16 BAR G

WHD SERIES TECHNICAL SPECIFICATION MATRIX

MODEL	CFM	m ³ /hr	Operating Pressure	Connection	Dimensions (mm)	Weight
WHD - 030	300 CFM	510	7–16 bar	1½" NB Flange	1450 × 1100 × 2500	450 kg
WHD - 050	500 CFM	850	7–16 bar	3" NB Flange	1650 × 1250 × 2300	720 kg
WHD - 075	750 CFM	1275	7–16 bar	3" NB Flange	1800 × 1300 × 2600	1120 kg
WHD - 100	1000 CFM	1700	7–16 bar	4" NB Flange	2000 × 1400 × 2700	1450 kg
WHD - 150	1500 CFM	2550	7–16 bar	5" NB Flange	2300 × 1600 × 2900	2100 kg
WHD - 200	2000 CFM	3400	7–16 bar	6" NB Flange	2600 × 1800 × 3100	2850 kg

Included Pre & After Filtration: Every WHD dryer includes pre-piped 0.01μ coalescing pre-filter and 1.0μ particulate after-filter to guarantee oil-free dry air and protect downstream processes.

THERMAL MANAGEMENT • WCP SERIES (2 TO 100 TR)

Industrial Process Water Chillers

High-Efficiency Air & Water-Cooled Chillers with Integrated SS304 Buffer Reservoir & Delivery Pumps

The WCP Series delivers precision process liquid chilling from +7°C to +25°C (down to -10°C with glycol). Complete package includes hermetic scroll / semi-hermetic screw compressors, insulated SS304 reservoir, stainless centrifugal circulating pump, and micro-controller.

2 TO 100 TR CAPACITY

SCROLL & SCREW

SS304 BUFFER TANK

±0.5°C PRECISION



WCP SERIES TECHNICAL SPECIFICATION MATRIX

MODEL	Tonnage (TR)	kW Capacity	Flow (LPM)	Tank (L)	Refrigerant	Dimensions (mm)
WCP 002	2.0 TR	7.0 kW	40 LPM	100 L	R407C	750 × 600 × 1050
WCP 005	5.0 TR	17.5 kW	100 LPM	200 L	R407C	1000 × 750 × 1250
WCP 010	10.0 TR	35.0 kW	200 LPM	350 L	R407C	1350 × 900 × 1450
WCP 020	20.0 TR	70.0 kW	400 LPM	600 L	R407C	1750 × 1100 × 1650
WCP 030	30.0 TR	105.0 kW	600 LPM	800 L	R407C	2200 × 1300 × 1850
WCP 050	50.0 TR	175.0 kW	1000 LPM	1200 L	R407C	2800 × 1500 × 2050
WCP 100	100.0 TR	350.0 kW	2000 LPM	2000 L	R407C	4200 × 1800 × 2300

Integrated Hydronic Package: No external water piping or field balance tank needed. Built-in SS304 reservoir acts as a thermal flywheel, eliminating compressor short cycling under variable mold or tool cooling demands.

SPECIALIZED PROCESS COOLING • CUSTOM BUILT

Specialized Industrial Chillers

Application-Specific Thermal Control: Anodizing, Medical Scan & Acid Cooling Systems

Anodizing Chillers (WAN)

Equipped with grade-1 pure Titanium immersion coils or shell-and-tube exchangers. Directly resists sulfuric, chromic, and oxalic acid baths in hardcoat anodizing lines.

Medical Scan Chillers (WMS)

Designed for MRI, CT, and linear accelerator cooling. Dual independent refrigeration circuits (N+1 redundancy) and automatic auto-failover prevent scanner downtime.

Acid Cooling Chillers (WAC)

Manufactured with Hastelloy C-276 or PTFE fluoropolymer heat exchangers and seal-less magnetic drive pumps to chill hydrochloric, nitric, and aggressive pickling fluids.

SPECIALIZED CHILLERS TECHNICAL MATRIX

MODEL SERIES	Capacity Range	kW Range	Wetted Metallurgy	Temp Control	Primary Applications
WAN Series	3 to 60 TR	10.5 to 210 kW	Pure Titanium Gr. 1	+15°C to +20°C	Aluminum Anodizing, Hardcoat Plating
WMS Series	3 to 30 TR	10.5 to 105 kW	SS316L / Copper	+8°C to +15°C (±0.2°C)	Hospital MRI / CT Scanners, Linear Acc.
WAC Series	2 to 50 TR	7.0 to 175 kW	Hastelloy C276 / PTFE	-5°C to +15°C	Chemical Reactors, Pickling Acid Tanks

Custom Metallurgy & Hazardous Area Options: Win Equipments engineers custom chillers with explosion-proof Zone 1 / Zone 2 electrical panels, flameproof motors, and remote condenser modules for chemical and offshore installations.

Thermal Systems • WFI Series (1 TO 30 TONS/DAY)

Industrial Ice Flake Machines

Continuous Production of Sub-Cooled Dry Ice Flakes (-7°C) for Concrete Batching, Fisheries & Chemicals

The WFI Series features stationary vertical chrome-plated or SS316 evaporator drums with internal rotating helical cutting blades. Produces dry, sub-cooled flake ice (1.5–2.2 mm thick) without mechanical scraping friction or refrigerant leakage risks.

1 TO 30 TONS/DAY

-7°C SUB-COOLED ICE

SUS304/SUS316 DRUM

CONTINUOUS 24/7 DUTY



WFI SERIES TECHNICAL SPECIFICATION MATRIX

MODEL	Daily Output	kg / 24h	kW Power	Compressor Make	Condenser Type	Dimensions (mm)
WFI 010	1 TPD	1,000 kg	4.8 kW	Danfoss (5 HP)	Air / Water Cooled	1350 × 1050 × 980
WFI 020	2 TPD	2,000 kg	8.5 kW	Bitzer Semi-Hermetic	Water Cooled Shell & Tube	1550 × 1200 × 1150
WFI 050	5 TPD	5,000 kg	19.5 kW	Bitzer Semi-Hermetic	Water Cooled Shell & Tube	2100 × 1500 × 1550
WFI 100	10 TPD	10,000 kg	38.0 kW	Bitzer Semi-Hermetic	Water Cooled Shell & Tube	2600 × 1800 × 1850
WFI 200	20 TPD	20,000 kg	74.0 kW	Bitzer Screw / Twin Recip	Water Cooled Shell & Tube	3600 × 2200 × 2300
WFI 300	30 TPD	30,000 kg	110.0 kW	Bitzer / Hanbell Screw	Water Cooled Shell & Tube	4200 × 2400 × 2600

Turnkey Ice Plant Systems: Win Equipments delivers fully integrated ice plants complete with automatic rake discharging systems, screw conveyors, sub-zero storage rooms, and evaporative condensing towers.

HEAT REJECTION • WCT SERIES (10 TO 1500 TR)

FRP Induced Draft Cooling Towers

Round Bottle & Square Crossflow FRP Towers Engineered for Efficient Industrial Evaporative Cooling

The WCT Series utilizes premium isophthalic polyester resin reinforced with chopped strand fiberglass matting. Features self-rotating low-pressure aluminum/brass sprinkler heads and non-clogging virgin PVC honeycomb fills with integrated drift eliminators.

10 TO 1500 TR CTI STANDARD BASELINE VIRGIN PVC FILLS CORROSION-PROOF FRP



WCT SERIES TECHNICAL SPECIFICATION MATRIX

MODEL	Tonnage (TR)	Flow (m ³ /hr)	Fan kW	Dia × H (mm)	Operating Wt
WCT 010	10 TR	7.8 m ³ /h	0.37 kW	Ø 850 × 1450	160 kg
WCT 025	25 TR	19.5 m ³ /h	0.75 kW	Ø 1250 × 1750	340 kg
WCT 050	50 TR	39.0 m ³ /h	1.50 kW	Ø 1650 × 2100	680 kg
WCT 100	100 TR	78.0 m ³ /h	3.70 kW	Ø 2250 × 2650	1400 kg
WCT 200	200 TR	156.0 m ³ /h	5.50 kW	Ø 3100 × 3200	2800 kg
WCT 500	500 TR	390.0 m ³ /h	15.0 kW	Ø 4800 × 4200	6800 kg

CTI Standard Thermal Ratings: Standard design conditions: Water inlet 42°C, Water outlet 32°C, Design wet bulb 28°C. Modular square configurations available up to 1500 TR with multi-cell distribution.

HEAT REJECTION • WCC SERIES (10 TO 500 TR)

Closed Circuit Coil Cooling Towers

Evaporative Closed-Loop Cooling Eliminating Process Water Contamination & Scale Formation

The WCC Series circulates internal process fluid through heavy-duty hot-dip galvanized or SS304 serpentine tubes. An external recirculating water deluge system combined with high-volume induced-draft fans rejects heat via evaporative sensible cooling with zero internal scaling.



10 TO 500 TR

CLOSED LOOP CIRCUIT

HDG / SS304 COILS

ZERO PROCESS FOULING

WCC SERIES TECHNICAL SPECIFICATION MATRIX

MODEL	Tonnage (TR)	Flow (m ³ /hr)	Fan kW	Spray Pump kW	Dimensions (mm)
WCC 010	10 TR	7.8 m ³ /h	0.75 kW	0.37 kW	1100 × 950 × 1850
WCC 025	25 TR	19.5 m ³ /h	1.50 kW	0.75 kW	1500 × 1200 × 2200
WCC 050	50 TR	39.0 m ³ /h	3.00 kW	1.50 kW	2100 × 1500 × 2650
WCC 100	100 TR	78.0 m ³ /h	5.50 kW	2.20 kW	2800 × 1800 × 3100
WCC 200	200 TR	156.0 m ³ /h	11.0 kW	4.00 kW	3600 × 2200 × 3650

Ideal for Sensitive Clean Equipment: Protects induction furnace coils, continuous casting molds, laser cutting resonators, and vacuum furnaces from limestone clogging and galvanic corrosion.

AUXILIARY AIR SYSTEMS • WRV & WADV SERIES

Air Receivers & Automatic Drain Valves

Boiler-Quality Carbon Steel Buffer Vessels & Electronic Zero Air Loss Condensate Drains

Air Receiver Tanks (WRV Series)

Manufactured in strict accordance with IS 2825 & ASME Sec VIII Div 1. Eliminates compressor pulsating cycles, stores energy during surge demand, and separates bulk moisture.

250 TO 5000 LITERS

7 TO 16 BAR

1.5x HYDRO TESTED

Zero Air Loss Drain Valves (WADV)

Capacitive liquid level sensing discharges accumulated moisture without blowing compressed air. Replaces lossy timer solenoid drains and prevents annual energy losses over ₹40,000 per valve.

ZERO AIR LOSS

0.8 TO 16 BAR

SELF-CLEANING

WRV AIR RECEIVER SIZING CHART

MODEL	Capacity (L)	Working bar	Test bar	Outer Dia (mm)	Height (mm)	In/Out Port
WRV 025	250 Liters	7 / 10 bar	12 / 17 bar	490 mm	1620 mm	3/4" BSP
WRV 050	500 Liters	7 / 10 bar	12 / 17 bar	630 mm	2000 mm	1" BSP
WRV 100	1,000 Liters	7 / 10 bar	12 / 17 bar	800 mm	2350 mm	1.5" BSP
WRV 200	2,000 Liters	7 / 10 bar	12 / 17 bar	1100 mm	2750 mm	2" Flange
WRV 500	5,000 Liters	7 / 10 bar	12 / 17 bar	1600 mm	3400 mm	4" Flange

Safety & Compliance Fittings: Every receiver is dispatched with an ASME/IS stamped safety relief valve, glycerin dial pressure gauge, manual drain ball valve, and factory hydro-test certificate.

AIR TREATMENT • WMF SERIES (20 TO 2000 CFM)

High-Efficiency In-Line Microfilters

Multi-Grade Coalescing, Particulate & Activated Carbon Vapor Filters (ISO 8573-1)

Grade P (Pre-Filter)

Particle removal down to 3.0 microns. Removes pipe scale, bulk rust, and coarse liquid water droplets.

Grade X (Coalescing)

Particle removal down to 1.0 micron. Liquid oil aerosol removal down to 0.1 mg/m³ (0.1 ppm).

Grade Y (Fine Coalescing)

High-efficiency fine sub-micron filtration to 0.01 micron. Residual oil content < 0.01 mg/m³.

Grade A (Carbon Vapor)

Activated carbon bed removing oil vapors and hydrocarbon odors down to 0.003 mg/m³ for food & pharma.

WMF SERIES TECHNICAL SIZING MATRIX

MODEL	CFM	Flow (m ³ /hr)	Port Connection	Max bar	Filter Element Qty
WMF 020	20 CFM	34	1/2" BSP	16 bar	1 × WE-020
WMF 060	60 CFM	102	3/4" BSP	16 bar	1 × WE-060
WMF 100	100 CFM	170	1" BSP	16 bar	1 × WE-100
WMF 200	200 CFM	340	1.5" BSP	16 bar	1 × WE-200
WMF 300	300 CFM	510	2" BSP	16 bar	1 × WE-300
WMF 500	500 CFM	850	2.5" BSP	16 bar	1 × WE-500
WMF 1000	1000 CFM	1700	3" Flange	16 bar	2 × WE-500
WMF 2000	2000 CFM	3400	4" Flange	16 bar	4 × WE-500

Differential Pressure Monitoring: Standard with top-mounted dual-color differential pressure gauge indicating optimal element replacement point to eliminate unnecessary line pressure drop.

TURNKEY PLANT CAPABILITIES & QUALITY COMMITMENT

Comprehensive Factory Testing & Lifecycle Engineering

Every piece of equipment manufactured at our Arasur works undergoes a rigorous 4-stage quality assurance protocol before dispatch:

Hydrostatic Proof Test

Every pressure vessel and coil assembly is hydro-tested to 1.5× maximum allowable design working pressure for a minimum 60-minute dwell period.

Helium Mass Spectrometry

All copper and stainless refrigerant circuits undergo vacuum helium mass spectrometry leak detection ensuring zero refrigerant loss across machine lifespan.

Full Load Performance Run

Continuous 8-hour live calorimeter trial verifying compressor motor draw, cooling capacity, cycle timing, and control logic safety interlocks.

Piping & Ducting Layouts

Custom 3D CAD modeling and compressor room piping layout engineering to ensure minimal line friction losses and optimal ventilation airflow.

Genuine Spares Availability

Over ₹50 Lakhs of original spares in stock at our Coimbatore facility including desiccant media, filter elements, contactors, and expansion valves.

Pan-India Field Engineers

Dedicated factory-trained mobile service technicians available for rapid commissioning, annual maintenance contracts (AMC), and emergency repair.

Contact Plant Engineering & Sales

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